

Block I—Topic 2: Land Use Regulation



Where we are...

- Urban Economics

- ▶ The study of the location choices of households and firms **within** an urban area and **institutional setting**

- This lecture: Focus on **land use regulation**

Contents

1. What is Land Use Regulation?
2. Welfare Economic Justifications for LUR: Correcting Market Failure
3. The Political Economy and the Origins of Land Use Regulation: Theory and Evidence
4. Empirical Evidence: External Benefits and Costs
5. Empirical Evidence: Net Welfare Impact

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What is land use regulation?

- Goes under many names: Planning, zoning, regulation
- Takes many forms
 - ▶ Horizontal (green belts)
 - ▶ Vertical (height restrictions, view corridors)
 - ▶ Constraints on alteration (historic preservation)
 - ▶ Zoning incl. 'exclusionary zoning' (MLSR)
 - ▶ Planning permissions
 - ▶ Density controls
 - ▶ Building codes

Three types of planning systems

1. **Master Plan** (e.g., France; rule based)

- ▶ Identifies land for urban use
- ▶ Plan identifies uses for all urban parcels
- ▶ Developer can build if in accordance with plans

2. **Zoning** (e.g., US; rule based)

- ▶ Municipalities can draw up zoning plans
- ▶ Zoning plans identify (i) what can be built in particular zones and (ii) restrictions (e.g., densities)
- ▶ Developers can build subject to building codes

Three types of planning systems—*cont.*

3. **Development control** (UK, some other Commonwealth countries; idiosyncratic)

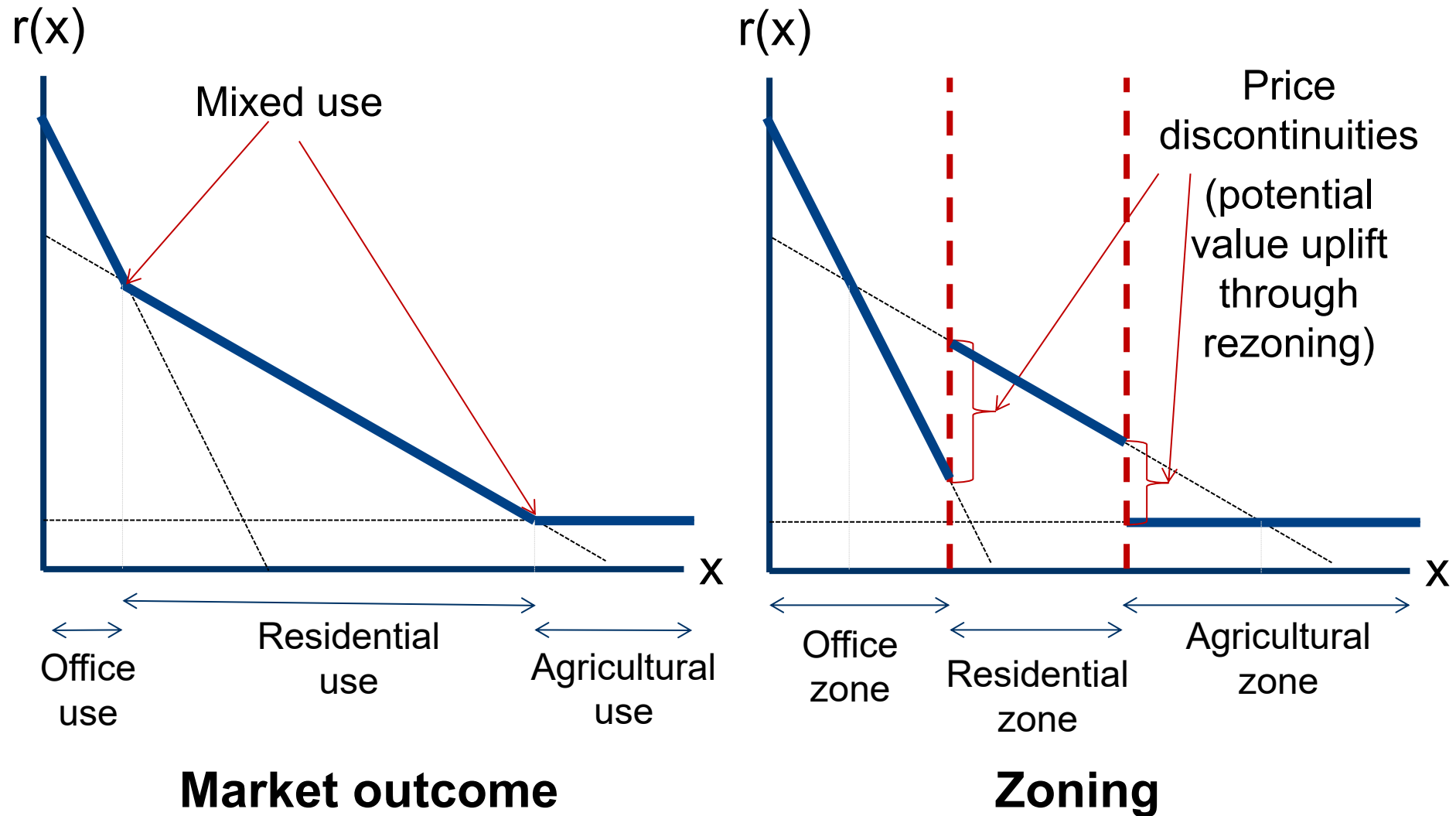
- ▶ Expropriates development rights of landowners
- ▶ Designates 'use' classes, whereas any change of 'use' requires development control permission (usually granted at local level)
- ▶ Aim is 'development control' or 'containment'

⇒ Other countries?

Land use planning is fundamentally an economic activity...

- Allocates a scarce resource—land—and defines property rights
- **But** without reference to prices
- Planning determines what can be built & where, **but** markets still set prices
 - ▶ Discontinuities in bid rent curve

Impact of zoning on bid rents



⇒ How can we analyse LUR?

Two theoretical approaches

- Normative view—What ought to be?
 - ▶ **Welfare economics view**
 - ▶ Asks: What is **optimal** form of land use regulation in presence of market failures?
- Positive view—What is? Why?
 - ▶ **Political economical view**
 - ▶ Asks: How can we explain **actual** outcome as result of political process?

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What market failures may justify land use regulation?

1. Technological externalities (Technological vs. pecuniary externality)

- ▶ *Examples:* Pollution, congestion, noise (negative); nice views on protected landmark buildings (positive)
- ▶ Instruments
 - Zoning: separates incompatible land uses
 - Containment and densification policies
 - Historic preservation (CAs; Listed Buildings)
 - View corridors & height restrictions

(Other non LUR-related: Pigou tax, carbon tax, congestion charge, emission trading, noise regulations)

Justifications for land use regulation—*cont.*

2. Public goods

- ▶ Goods that are non-rival in consumption and non-excludable
- ▶ *Examples:* Public parks, historic buildings, areas of outstanding natural beauty
- ▶ Instruments to prevent free riding:
 - Designation of land as urban parks, ‘habitats’, National Parks, wilderness areas or recreational areas, protection of open spaces
 - Historic urban preservation (Conservation Areas to protect cityscapes; Listed Buildings)

Justifications for land use regulation —cont.



3. Collective action problem

- ▶ Land assembly (combination of individual parcels of land for (re-)development often has potential to increase social welfare (via collective action)
- ▶ *Examples:* New infrastructure provision such as new roads or railway tracks; large scale buildings
- ▶ *Problem:* Each owner may seek to capture disproportionate share of potential surplus associated with land assembly (**strategic hold-out problem**)
- ▶ Instrument to solve problem: **Eminent domain**
 - Expropriation of private property for public use along with compensation

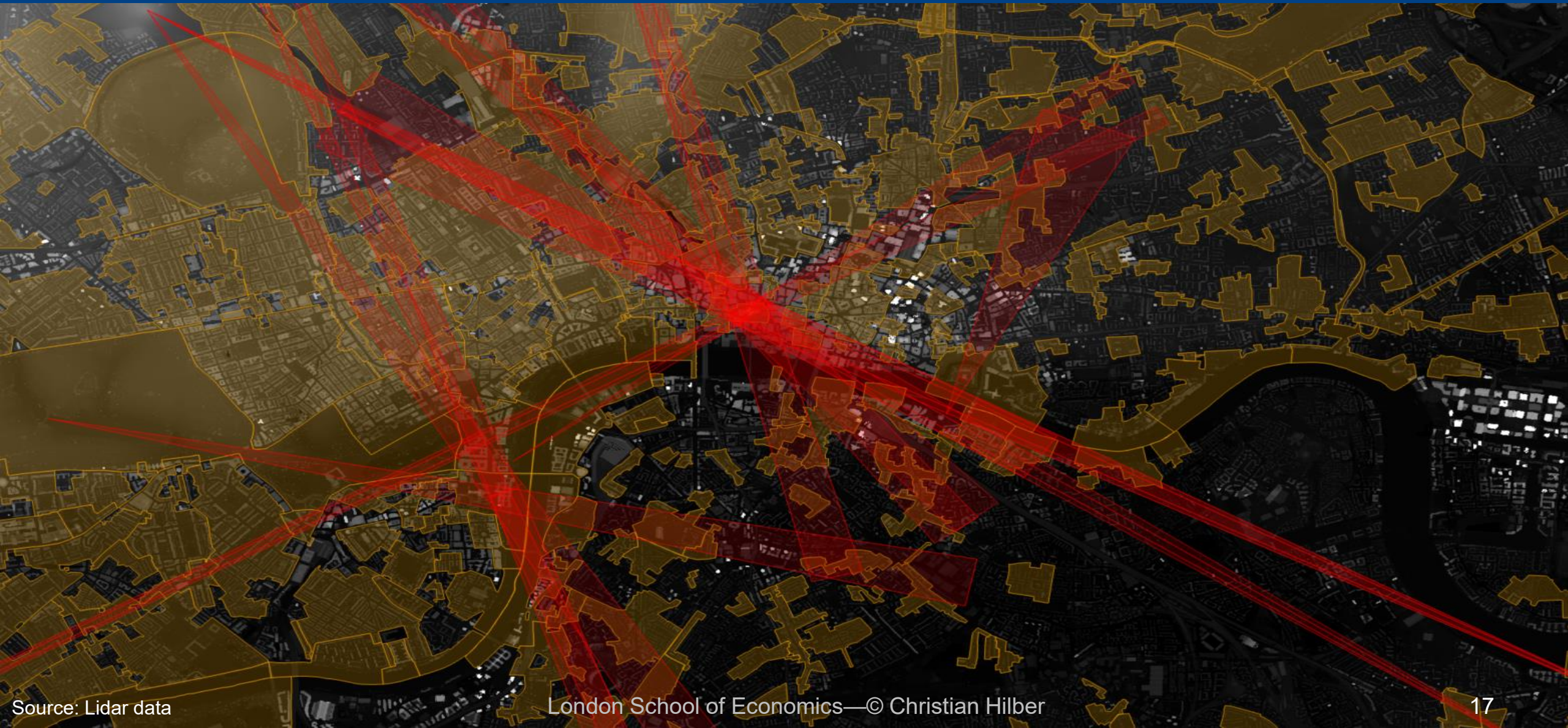


What is the 'problem' with welfare economics approach? (Discuss)

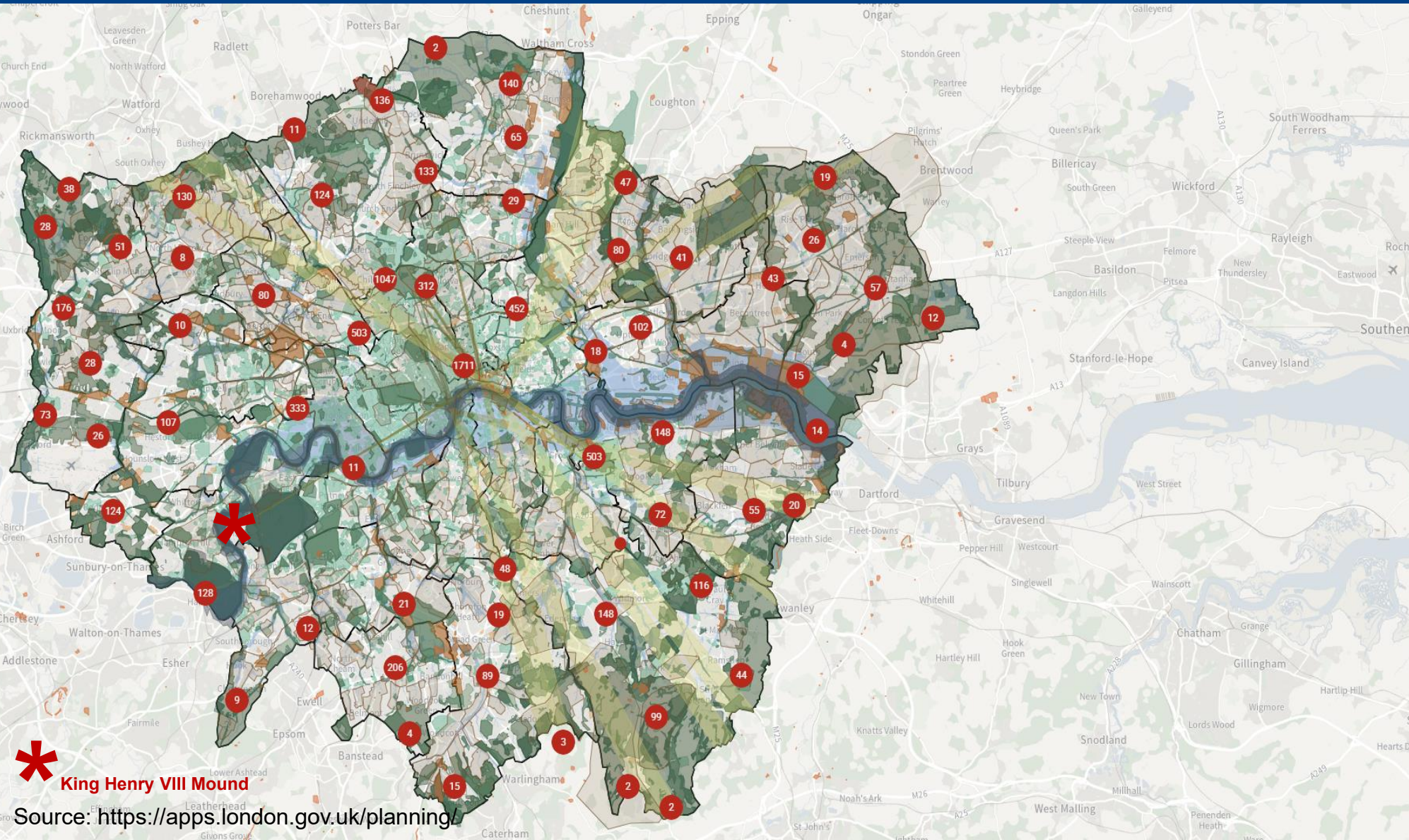
- Information problem
 - ▶ Difficult to quantify externalities & corrective actions
 - ▶ Difficult to determine optimal level of public good provision
- Decisions not made by 'benevolent planner'!
- ⇒ In reality: LURs = outcome of political process and vested interests...
- ⇒ Outcomes sometimes seem far from welfare maximizing...

Example: Central London

Red rays: Protected vistas; yellow areas: Conservation Areas



Planning restrictions in Greater London



Search near a location:
e.g. enter your postcode

Protection Good Growth Borough Layers Context

- Designated Open Space
Designated Open Space includes lands designated as Green Belt, Metropolitan Open Land (MOL) and other public open spaces... Read More
- Green Belt
- Metropolitan Open land
- Other Open Space
- Site of Importance for Nature Conservation (SINC)
- Sites of Special Scientific Interest (SSSI)
- Conservation Areas
- Ancient Woodland
- Strategic Industrial Locations (SILs)
- Locally Significant Industrial Sites (LSIS)
- Flood Risk
- Safeguarded Wharves
- Thames Policy Area
- Listed Buildings
- Protected Vistas
- Scheduled Monuments
- Registered Battlefield
- Critical Drainage Areas

Find Layers & Sites by Location

Begin Radius Search

* King Henry VIII Mound

Source: <https://apps.london.gov.uk/planning/>

* Example: Protected view from King Henry VIII's Mound (Richmond Park)—*created in 1710*



... who are very vocal ...



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Who are the relevant agents in 'political economy game' (Discuss)

- **Homeowners (NIMBYs)**
 - ▶ Have incentives to protect asset values + community composition & finances (**vote**)
- **Renters (tenants)**
 - ▶ Regulation has pros and cons (**vote**)
- **Landlords (= those who rent out)**
 - ▶ Have incentives to protect asset values (**lobby**)
- **Owners of undeveloped land (w/o permission)**
 - ▶ Prefer flexible planning system (**lobby**)
- **Developers**
 - ▶ Depends on whether they have planning permission for site already (**lobby**)

Two political economy views

- **Homevoter hypothesis** (Fischel 2000, 2001)
 - ▶ Homevoters = local homeowners who vote to protect asset values
 - ▶ Regulatory restrictiveness = local voting outcome of homeowners vs. renters
 - ▶ Prediction: homeownership $\uparrow \Rightarrow$ restrictiveness $\uparrow$
- **Influential landowner hypothesis** (Hilber & Robert-Nicoud 2013)
 - ▶ Voting and lobbying relevant ('one \$ one vote')
 - ▶ Restrictiveness = outcome of power struggle between owners of developed land (homeowners, landlords) vs. owners of undeveloped land (& renters)
 - ▶ Prediction: Physical development $\uparrow \Rightarrow$ restrictiveness $\uparrow$

Evidence in favor of two theories

■ Homevoter-hypothesis

- ▶ Anecdotic evidence: ‘homevoters’ behave as NIMBYs
- ▶ [Dehring et al. \(2008\)](#) exploit voting data on local referendum in Texas & find evidence in support
- ▶ [Greaney et al. \(2025\)](#): homeownership $\uparrow \rightarrow$ regulation gets tighter & construction $\downarrow$

■ Influential Landowner-hypothesis

- ▶ Historical evidence: most desirable places were developed & regulated first
- ▶ [Hilber and Robert-Nicoud \(2013\)](#) explore cross-sectional variation in ‘degree of development’ across US metro areas (using IV) & find evidence in support
- ▶ [Sole-Olle & Viladecans-Marsal \(2012\)](#): lobbying important in explaining land supply in Spain

Other evidence

- **Political ideology matters too...**

- ▶ [Kahn \(2011\)](#): ‘Liberal’ cities in US (with Democratic majority) are *more* restrictive
 - But also: [Ferreira & Gyourko \(2023\)](#): Political partisanship does not affect housing supply, suggesting US cities follow US voter
- ▶ [Sole-Olle & Viladecans-Marsal \(2013\)](#): Cities in Spain controlled by left-wing parties convert less land from rural to urban use
- ▶ [Hilber and Vermeulen \(2016\)](#), [Hilber & Mense \(2025\)](#), [Greaney *et al.* \(2025\)](#): Labour dominated LAs in England have lower refusal rates of planning applications

Preliminary conclusions

- LURs have potential to increase social welfare
 - **But** in reality: LURs not determined by ‘benevolent planner’ ⇒ rather outcome of political process
 - Two main channels: voting + lobbying
 - Outcome likely not welfare maximizing
- ⇒ Do benefits of LURs outweigh costs?
(Too much or too little?)

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Evidence of external benefits (focusing on preservation policies & employing DiD-approach)

- Been *et al.* (2016) – US
 - ▶ Designation has strong positive effect outside (12%)
 - ▶ Inside: effects are positive but only outside Manhattan
 - ▶ Generally: Positive effects are smaller in areas where option to build unrestricted is higher (value of option to redevelop)
- Koster *et al.* (2016) – NL
 - ▶ Premium for houses with views on designated buildings: 3.5%
 - ▶ Designated buildings do not trade at premium

And for the UK?

- Ahlfeldt *et al.* (2017) – UK
 - ▶ *Historic preservation designation* has no effects on prices inside CAs and no spillovers on adjacent areas in short run
 - ▶ Some evidence of weak positive spillover effects in long run
 - ▶ Authors' interpretation of “zero finding”:
 - “The policy is either deliberately adhering to the interests of **local owners** or [...] homeowners are able to influence the outcome of local policies successfully in their interest.”

Do positive price effects necessarily imply social welfare increase? – Why (not)? (Discuss)

1. **Costs & benefits are not always (fully) capitalized** into prices – Examples:
 - ▶ [C:] Preservation hinders or prevents altogether energy efficiency installations $\Rightarrow$ carbon footprint $\uparrow$ (not capitalized in HPs) $\Rightarrow$ social welfare $\downarrow$
 - ▶ [B:] Existence value of St. Paul's cathedral not reflected in higher nearby house prices
2. Even if there were full capitalization, it **only** suggests preservation is **optimal for marginal homebuyer**, but not (necessarily) welfare maximizing!

Do positive price effects necessarily imply social welfare increase? – Why (not)? (Discuss, cont.)

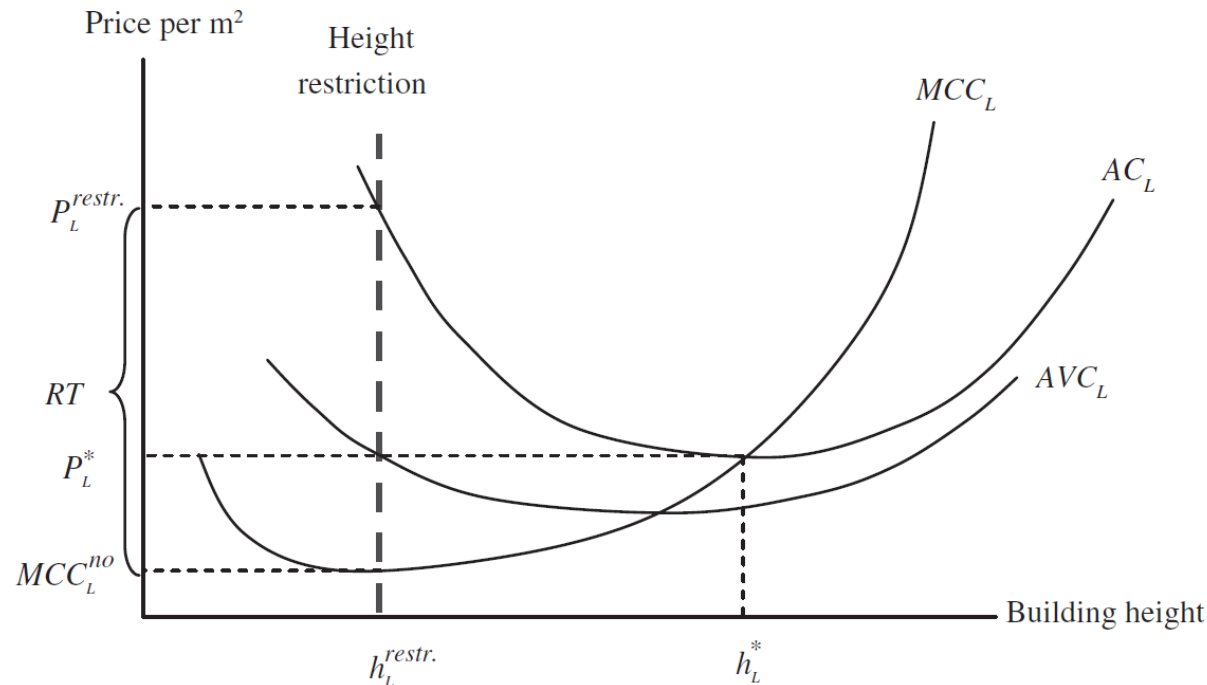
3. **DiD does not capture aggregate ‘market-wide’ (general equilibrium) effects**
- ▶ Preservation on wider scale (like in London) constrains housing supply at city level (can’t redevelop at higher density!) $\Rightarrow$ HP & rents at city level $\uparrow \Rightarrow$ individual housing consumption $\downarrow \Rightarrow$ social welfare $\downarrow$
 - ▶ HP & rent increases at city level do not represent a welfare gain but are effectively a cost or so-called ‘regulatory tax’ to consumers!
- $\Rightarrow$ Be careful interpreting evidence of positive price effects in DiD as ‘net welfare gain’ for society as a whole!

⇒ What about (other) costs of land use regulation?

LUR may drive up cost of housing & commercial space

- Glaeser *et al.* (2005) estimate ‘**regulatory tax**’ for residential high-rise buildings in Manhattan
 - ▶ >100% of construction costs (>50% of price)

- Idea:



Source:
Cheshire and
Hilber (2008)

Estimated 'regulatory tax' for office buildings *as % of construction costs, by city*

Location	Mean 1999-2005 (Cheshire & Hilber 2008)
London West End	809
London City	488
Frankfurt	437
Stockholm	379
Milan	309
Paris	305
Edinburgh	291
Barcelona	269
Birmingham	250
Manchester	230
Glasgow	204
Amsterdam	202
Leeds	193
Bristol	157
Brussels	68

Effects on vacancies and commuting distances...

- Cheshire *et al.* (2018)
 - ▶ Exploit variation in regulatory restrictiveness (refusal rates) across English LAs & over 30 years
 - ▶ 1 SD increase in refusal rates ↑ **vacancy rate** by 0.9%-points (23%) & causes **commuting distance** to ↑ by 6.1%

Adverse effects on growth and retail output

- Hsieh and Moretti (2019) + Duranton & Puga (2023) – for US
 - ▶ Quantify amount of **spatial misallocation of labor** due to stringent restrictions in high productivity cities such as NY or SF
 - ▶ H&M (2019): Calibrations of spatial equilibrium model suggest constraints lowered aggregate US growth by **36% b/w 1964-2009**
 - ▶ D&P (2023): Calibrations suggest relaxing constraints would increase aggregate output by **8.0%** and aggregate consumption by **2.2%**
- Cheshire *et al.* (2015) – for UK
 - ▶ Explore impact of ‘**Town Centre First**’ (TCF) policy on retail output using DiD
 - ▶ Policy imposed loss of output of **32% on representative store opening post-implementation**

Energy cost of preservation policies

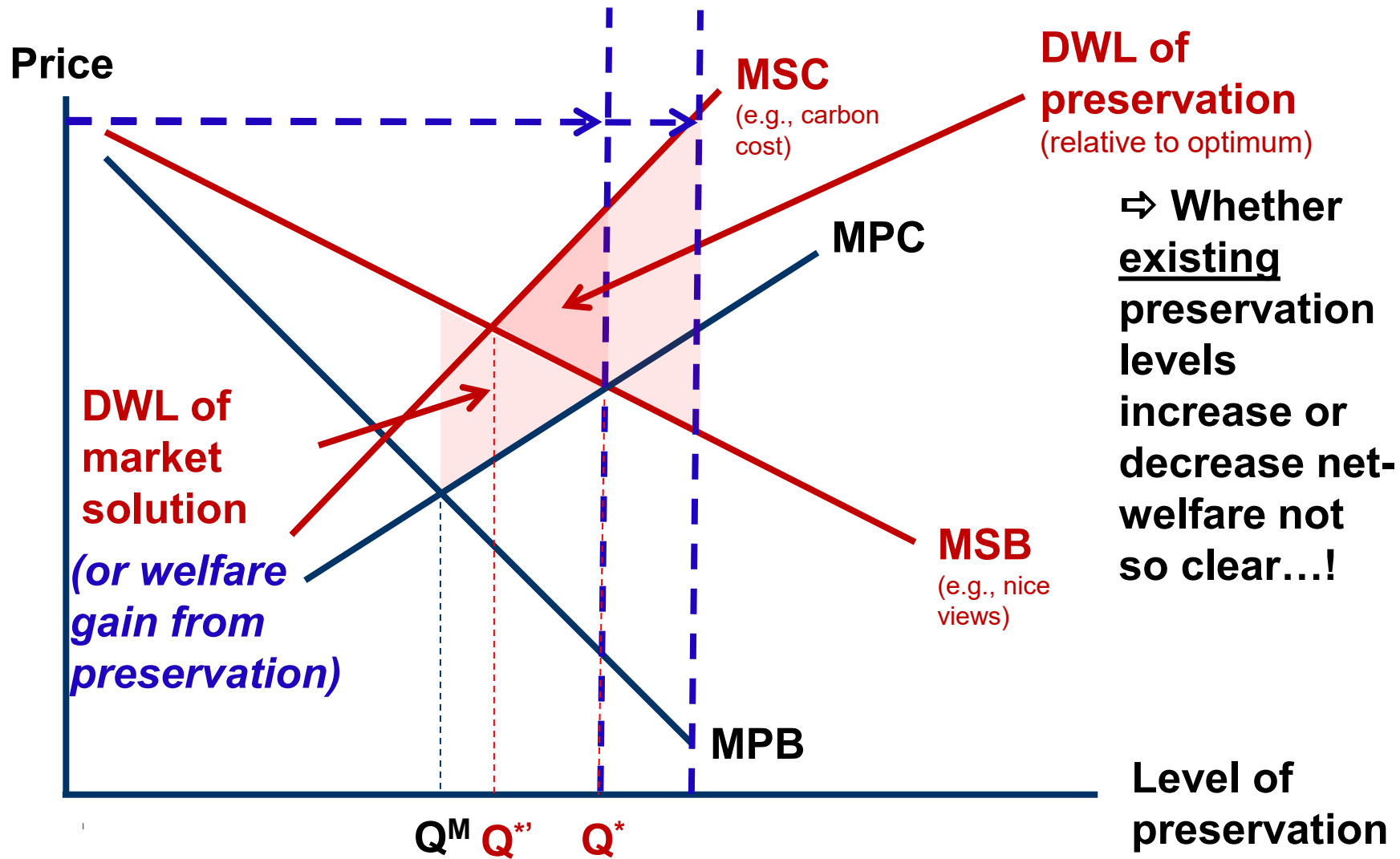
- Hilber *et al.* (2019)
 - ▶ Show that rising energy prices in England induce increase in energy efficiency installations & reduce energy consumption, **but** much less so in Conservation Areas (CAs) and Listed Buildings (LBs)
 - ▶ Preservation policies in England raise energy consumption & social carbon cost by **PV of £11.6k and £2.4k per dwelling**

What about **net** welfare effect?

- Land use regulation has benefits and costs
- Can we say anything about **net welfare effect**?
 - ▶ Is land use in e.g., US or UK too tightly regulated?
- ⇒ Depends on nature & magnitude of market failures & actual level of restrictiveness...

Theoretically ambiguous...

(Example: Historic preservation)



Evidence on net welfare effect?

- Difficult to quantify...
- Could look at net effect on property prices **BUT** some externalities are not capitalized, e.g.
 - ▶ Existence value of protecting St. Paul's cathedral
 - ▶ Carbon cost of preservation policies via raising cost of/hindering energy efficiency installations
- Some attempts to identify overall net welfare impact nevertheless...



Most rigorous effort to date: *Turner et al. (2014)*

- Idea: Exploit cross-border changes in regulation, development & prices in areas near municipal borders (boundary discontinuity)
- Decompose effects of regulation into three components
 1. **Own-lot effect:** Cost of constraints to the owner of parcel
 2. **External effect:** Value of constraints on one's neighbors
 3. **Supply effect:** Effect of regulated scarcity of developable land (aggregate effect)

Evidence of net welfare impact

- Finding: Estimates suggest large **negative net effect** of regulation on welfare
 - ▶ Mainly driven by adverse own-lot and external effects
- ⇒ LUR on average **too restrictive** in U.S.
(Qualitatively same finding for UK by [Cheshire & Sheppard, 2002](#))

Conclusion

- Land use planning has potential to improve social welfare via correcting market failures
- In reality: Planning not determined by ‘benevolent social planner’ but outcome of political process (voting + lobbying)
- Regulation has both external benefits and external costs
- Evidence is suggestive that (at least parts of) US & UK are too tightly regulated

Preparations for *Block I – Seminar 2* (Week 4)

- **Case study/debate with sub-group presentations and discussion** (building on Topic 2)
 - ▶ **Topic: Historic preservation**—(How much) should governments intervene in property markets to preserve historic building structures?
 - ▶ **Details:** See special handouts
 - ▶ **Relevant readings:** See special handouts

Preparations for Block II – **Seminar 3** (Week 5)

- **Problem set (case study) with sub-group presentations and discussion** (building on Topic 3)
 - ▶ **Topic:** Estimating the housing supply price elasticity for UK regions
 - ▶ **Details:** See special handout
 - ▶ **Relevant key readings:** See handout

Reflection

- Write down the 3 most important things that you have learnt so far on GY457...

Technological vs. pecuniary externalities

- **Technological externality:** An effect on a third party's production or utility that happens outside the price system, directly changing their inputs or output levels – affecting the third party's *ability* to produce or consume.
 - ▶ Justify policy intervention
 - ▶ Example: pollution harming production
- **Pecuniary externality:** An effect on a third party's profits or budget that occurs through changes in market prices (input costs or output prices) – affecting a third party's *profits* or *budget* but not their physical production capacity
 - ▶ A transfer, not a net loss; do not generally warrant policy intervention
 - ▶ Example: A big firm opening an office, increasing local house prices & rents